

Matheus Fonseca

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EDUCATION

CUNY Brooklyn College

Expected May 2029

Bachelor of Science, Major in Computer Science

- **Relevant Coursework:** Programming Using Java

CS50x: Introduction to Computer Science – Harvard/edX

Dec 2024

- Completed Harvard's introductory CS course covering **algorithms**, **data structures**, memory, and web development.
- Built hands-on projects using **C**, **Python**, **SQL**, and **HTML/CSS**, including a final web application.

EXPERIENCE

AlgoVerse

Remote

AI Researcher

Aug 2025 – Present

- Researching the breeding of LLMs through evolutionary techniques with the goal of increasing overall robustness.
- Currently engaged in ideation, literature review, and early-stage idea testing to identify feasibility.
- Participating in structured mentorship, weekly workshops, and collaborative sessions to strengthen understanding of AI concepts and research methodology.

Cava Group, Inc.

New York, NY

Team Member

Apr 2025 – Sep 2025

- Selected to work at a **CAVA NYC Learning Academy**, a high-performance training hub where general managers prepare to launch new stores across the U.S.
- Demonstrated adaptability across multiple fast-paced roles including line service, grill, prep, and dishwashing.
- Consistently recognized by management for exceptional work ethic, curiosity, and high-quality customer service.
- Recognized for promotion to managerial position but chose to leave to prioritize studies.
- Built strong relationships with managers and teammates, contributing to a positive and efficient store environment.

PROJECTS

SendToMac | python, supabase, rumps, swift

- Developed a **cross-platform application** enabling asynchronous transfer of text and URLs from iOS to macOS, even while the Mac is offline.
- Implemented **background macOS client** using **Python**, **AppKit**, and **rumps** to display real-time alerts with user actions (Open, Snooze, Ignore).
- Prototyped iOS interface using **SwiftUI** with AI assistance to accelerate design and implementation.
- Enabled content transfer via **REST API** with responsive success/failure alerts.
- Integrated scheduling and expiration logic using timestamps to manage message flow and cleanup.

Balatest (backend) | python, flask, pypdf2, gemini, prompt engineering

- **Led a 4-person team** from ideation to completion at **CCNY's Byte Hacks**, delivering a playable educational card game prototype in 24h earning positive feedback from judges/participants.
- Engineered **backend service** in **Flask** to parse PDF content with PyPDF2 and dynamically generate JSON-based playing cards across multiple difficulty levels.
- Integrated **Gemini API** with prompt engineering techniques for semantic processing, enabling adaptive gameplay mechanics and content generation.
- Collaborated on game logic and design to align backend functionality with front-end mechanics and user experience.

Chess Web-App | html, css, javascript

- Developed an interactive **chess web application** with full rules enforcement, including check, checkmate, stalemate, and legal move validation.
- Implemented real-time piece selection with visual highlights and move restrictions based on king safety and game rules.
- Built modular move logic and game state tracking to ensure accurate gameplay flow and turn handling.
- Designed a responsive, user-friendly interface using **HTML**, **CSS**, and **JavaScript**.

SKILLS

Technical: Python, Flask, C, JavaScript, CSS/HTML, SQL, Supabase, Git

Spoken Languages: English (Native), Portuguese (Native), Spanish (Basic)

Interests: Chess, Weightlifting, Swimming, Travel, AI, Movies